

Multi-Scale Analysis of Sea States in Nazaré: From Standard Swells to Extreme Event Drivers (2000–2020)

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1 Introduction

Portugal's Nazaré region is mostly known for its big wave phenomena. Its uniqueness is caused by a specific structure of the ocean floor. In the ocean bed, there is a massive canyon named Canhão da Nazaré, which is one of the deepest and most extensive underwater structures in the Northern Hemisphere. It acts as a natural amplifier for waves and, through interference phenomena, helps them reach heights of 15–30 meters in the winter season and 2–5 meters in summer. **The main question that was analyzed was the driving factors of the waves in Nazare and the climatic variables causing them with emphasis on extreme events.**

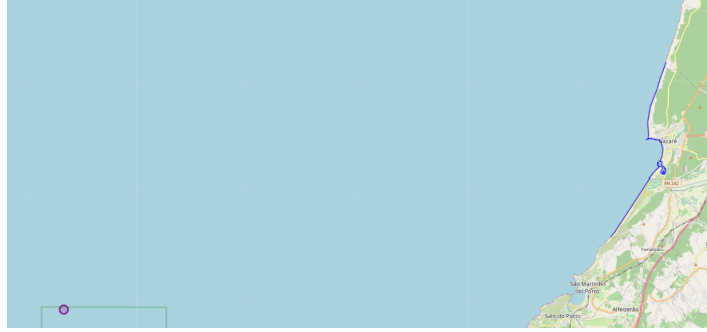


Figure 1: Spatial location of the ERA5 offshore grid point (39.5°N , 9.55°W) relative to the Nazaré coastline. Basemap: OpenStreetMap via Folium.



Figure 2: Visualization of the Nazaré Canyon (*Canhão da Nazaré*) seafloor topography, highlighting the structures responsible for wave amplification. Image credit: National Geographic.

2 Data and Plan

In this paper, an analysis of Nazaré waves was conducted at an offshore ocean point located at latitude 39.5 and longitude -9.55 . The dataset was sourced from the Copernicus Climate Data Store and is based on the ERA5 reanalysis. The dataset covers wave heights ranging from 0.48 m to 9.49 m. At this location, the wave heights are measured prior to the influence of the Nazaré Canyon; the average amplification factor provided by the canyon is approximately $2.0\times$ to $3.0\times$ relative to the offshore wave height. This point was selected due to its open-ocean characteristics, which minimize the influence of land-based atmospheric signals to which ERA5 can be susceptible. The analysis of Nazaré waves is divided into four sections, as follows:

1. Descriptive Analysis
2. Diagnostic Analysis
3. Seasonal Analysis
4. Predictive Analysis

2.1 Dataset and Variables

Variables used for further analysis of Nazare waves have been concluded into Table 1.

Variable	Unit	Description
Significant Wave Height (H_s)	m	Predictive target variable (range: 0.48 – 9.49) of wave height
Peak Wave Period (T_p)	s	Period of the most energetic wave components
Mean Wave Period (T_m)	s	Average period of all wave components in the spectrum
Wind Speed (U_{10} V_{10})	km/h	Derived from u and v components of wind
Wind Direction	Cardinal	8 cardinal sectors (N to NW) via One-Hot Encoding
Sea Level Pressure	hPa	Barometric pressure at mean sea level
Sea Surface Temp (SST)	$^{\circ}\text{C}$	Temperature of the ocean surface layer
Surface Heat Flux	W/m^2	Net energy exchange; negative indicates ocean cooling
Month	-	Seasonal indicator (1 – 12) used for seasonality
Year	-	Temporal indicator (2000 – 2019) for long-term trends

Table 1: Summary of meteorological, oceanographic, and temporal features used in the analysis.

2.2 Pre-processing

The pre-processing of the dataset can be divided into three major steps, as follows:

1. **Construction of a virtual point:** ERA5 provides data on the grid through 0.25° steps. The first downloaded dataset was of latitude 39.50 and two longitude points of -9.80 and -9.30. These points were then constructed into one virtual point using mean and groupby methods in the Python programming language. The choice of constructing the point was taken because of the specific nature of climatic data.
2. **Unit conversion:** The reconstructed dataset was then subjected to the task of unit conversion. The primary units of measurement used in the Copernicus System were converted into more analysis-based, such as:
 - Kelvin to Celsius
 - Pascal to hectopascal
 - Wind speed (km/h), computed from the $u10$ and $v10$ components
3. **One-hot encoding** Computing of wind direction from $u10$ and $v10$ components using the arctan function. The resulting angles were classified into eight directional sectors (N, NE, E, SE, S, SW, W, NW). One-hot encoding was then applied to allow the use of wind direction in correlation analyses of peak wave heights.

3 Descriptive Analysis

3.1 Initial data distribution

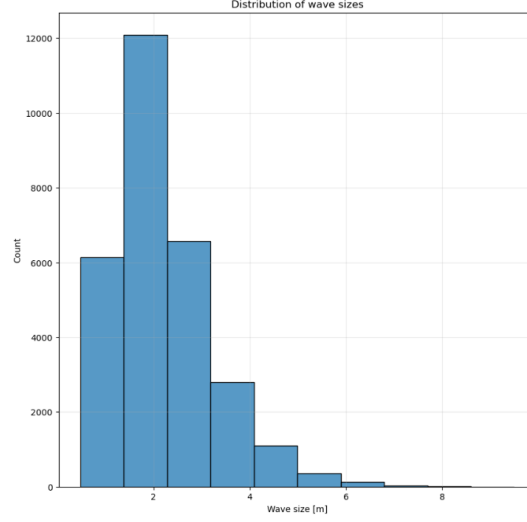


Figure 3: Frequency distribution of significant wave height (H_s) at the selected point.

The histogram shows the distribution of wave heights for the period from 1 January 2000 to 31 December 2020. It exhibits a strong positive skewness, as expected for wave height distributions.

3.2 Variables correlation

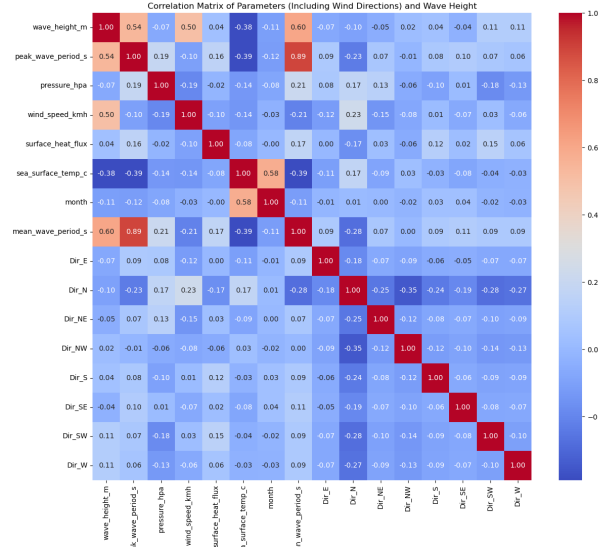


Figure 4: Matrix of Pearson linear correlation coefficients with variables describing wave height

The heatmap presented in **Figure 4** shows relatively high correlations between **significant wave height** and both the **mean** and **peak wave periods**, with correlation coefficients in the range of approximately 0.50–0.60. A similarly relationship is observed for wind speed, which shows high correlation values in case of wave height. These results indicate that the primary drivers of wave conditions at Nazaré are, as anticipated, swell characteristics and wind speed. Wind direction variables exhibit generally low linear correlations with the exception of SW, W and N which follow the direction of the waves towards the coast of Nazaré.

3.3 Scatterplots of the physical waves drivers

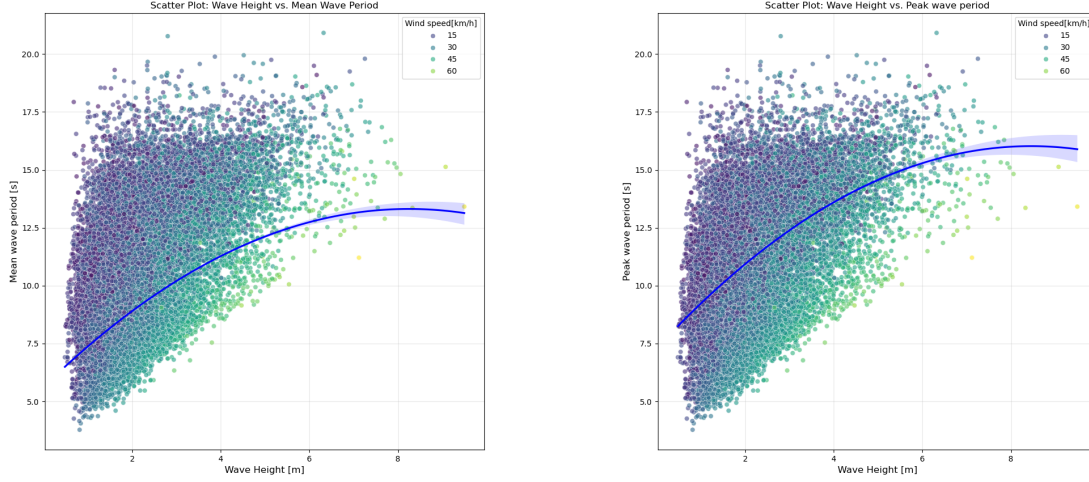


Figure 5: Relationship between Significant Wave Height (H_s) - Mean and Peak Wave Periods, with color shading representing Wind Speed with added regression curve of 2 degree.

The graph shows upward trend in wave height depending on the variables with the highest correlation. The most notable feature of the plot is near the end of the regression line, in both cases the curve starts to decrease in value which can mean a change in the factors causing extreme waves and will be further discussed in section 2 **Diagnostic Analysis**.

4 Diagnostic Analysis

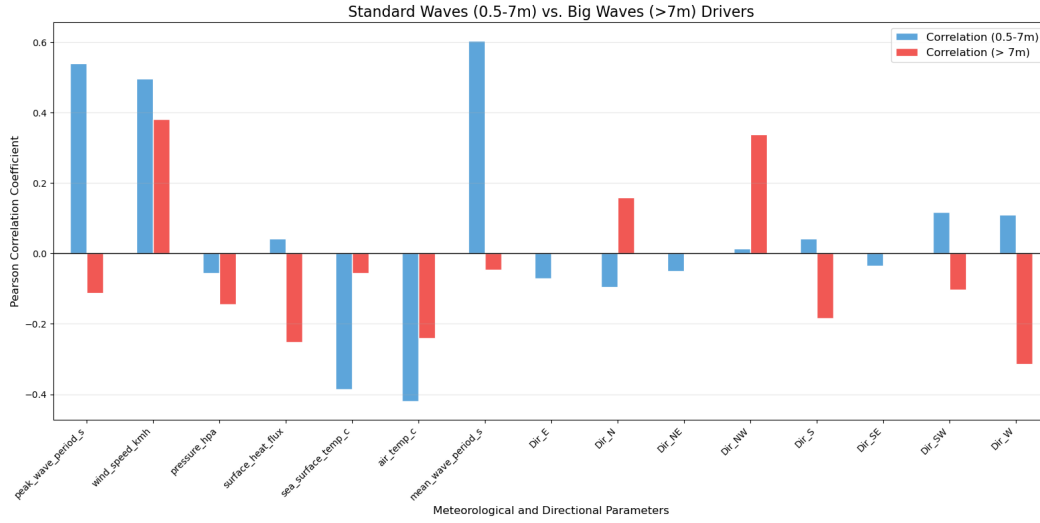


Figure 6: Change of correlation from normal to Big waves.

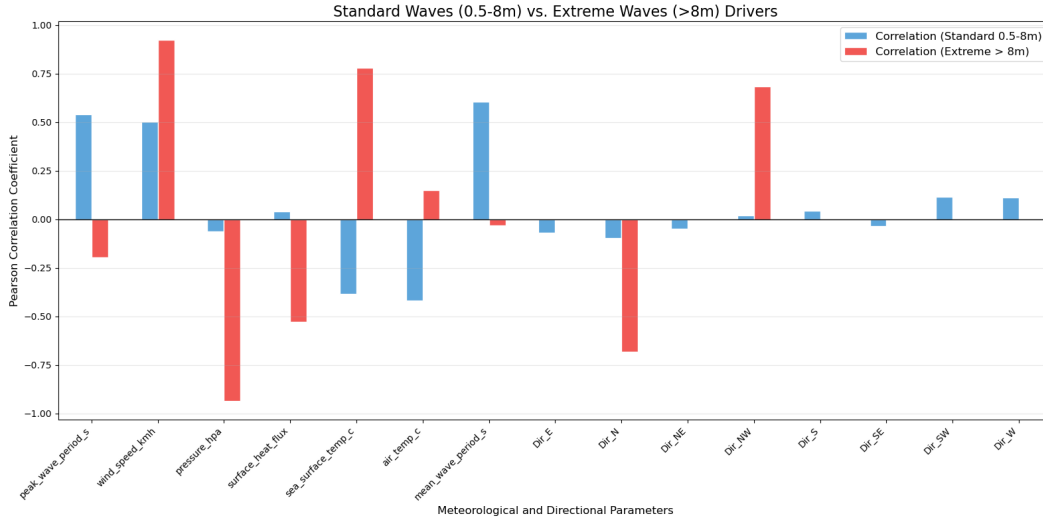


Figure 7: Change of correlation from normal/big to Extreme waves.

Figure 6 and Figure 7 both represent a shift in variables causing extreme and big waves. The plot was divided into two because of a better representation of correlation. In the 21-year dataset there were only 4 events when a recorded peak wave exceeded 8m. Which bring calculations dangerously close to small-sample uncertainty. The seven-meter mark works as an intermediate point between correlation changes and shows better mathematical results due to the larger size of the sample.

The results confirm that the initial assumptions derived from the diagnostic analysis are valid. The plots reveal a clear shift in the dominant drivers of large and extreme wave events. This shift is most evident in the behavior of sea level pressure and sea surface temperature. For extreme wave conditions, a pronounced decrease in atmospheric pressure and an increase in sea surface temperature appear to play a more significant role than the mean and peak wave periods.

An additional noteworthy result concerns the role of wind direction. The subset of waves exceeding 7 m exhibits a broader sensitivity to wind direction, whereas extreme events exceeding 8 m are strongly associated with northwesterly winds. In contrast, the 7 m subset also shows a substantial contribution from northerly winds, which contributes to wave amplification.

5 Seasonal Analysis

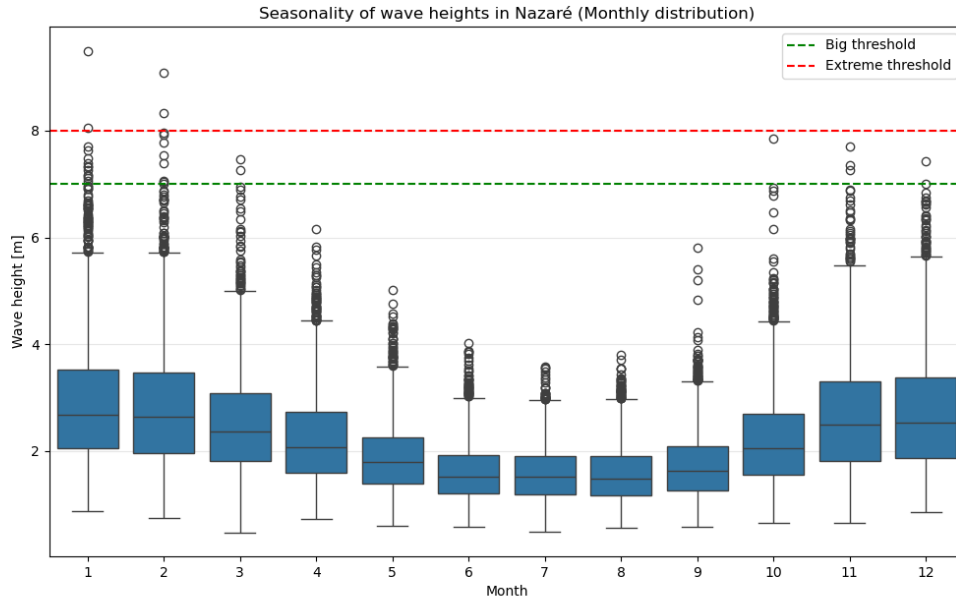


Figure 8: Monthly seasonality of wave phenomena in Nazaré

The box plot illustrates the monthly seasonality of wave height at Nazaré. A clear predominance of higher wave heights is observed during the winter months. It is common due to the winter nature of the Atlantic Ocean, which is characterized by stronger storms, deeper low-pressure systems and changes in air flow. All of that leads to big wave forms appearing more frequently.

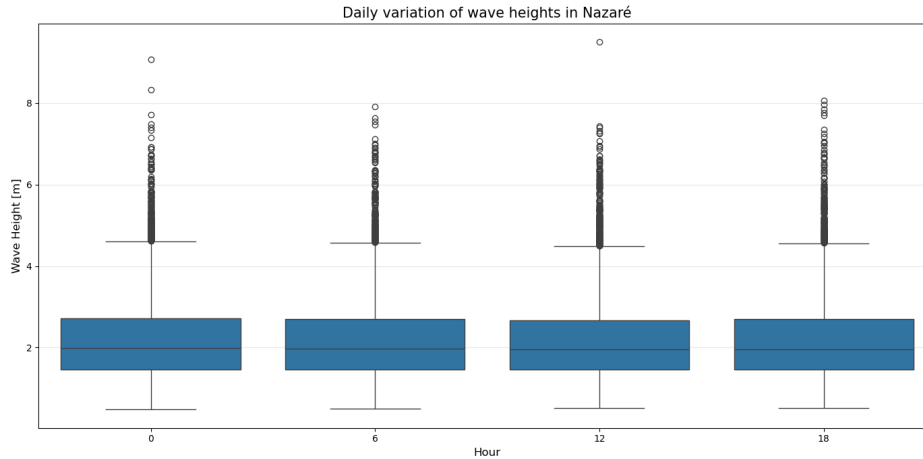


Figure 9: Daily seasonality of wave phenomena in Nazare

On the contrary to **Figure 8**, the graph showing daily seasonality does appear to have no distinguishing features. This supports the hypothesis that big waves in Nazare are caused by pressure changes and swell rather than breeze or other variables related to the rhythm of the day.

6 Predictive Analysis

The final part of the analysis was based on the application of the XGBoost algorithm to predict wave heights. Such an approach may serve as an initial component of a larger system architecture, for example an early warning system for extreme wave events.

The first tests with larger coefficients were successful with:

- Coefficient of Determination (R^2 score): 0.89
- Mean Absolute Error (MAE): 0.24m

Although such an architecture would require further refinement before being applied in real-world scenarios, the results indicate that machine learning methods can effectively support the prediction of climate-related variables.

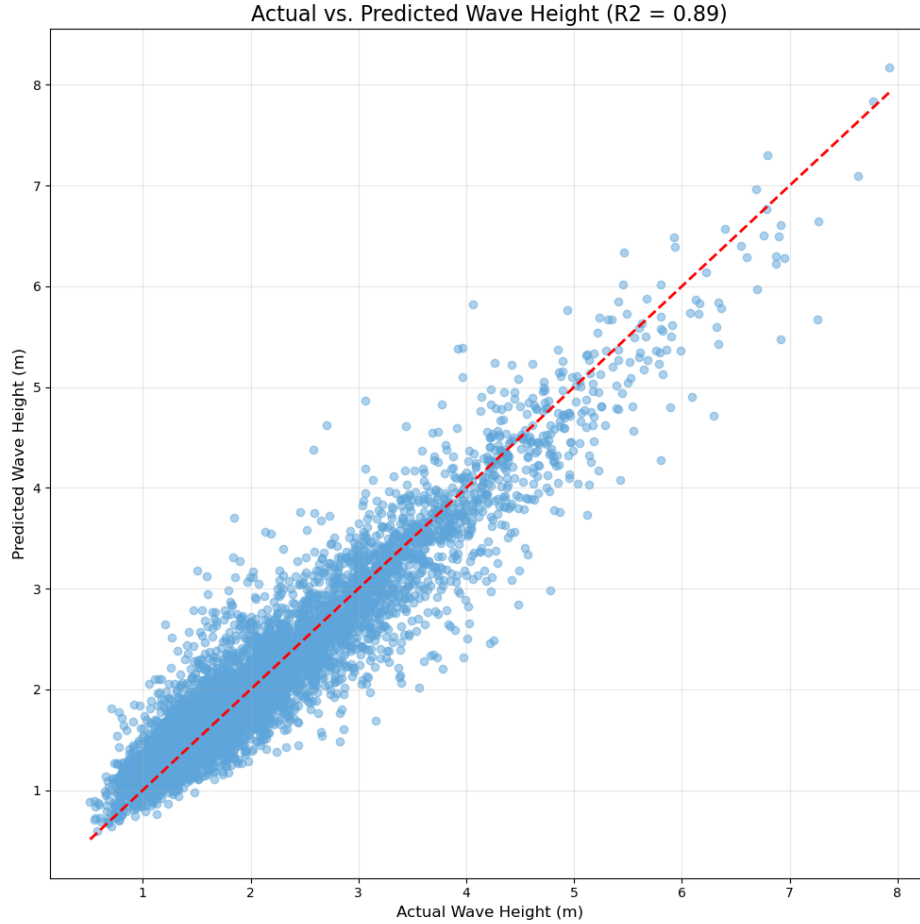


Figure 10: Values of wave heights predicted with XGBoost with added regression line

As shown in Figure 10, the XGBoost algorithm is able to capture the relationship between the input variables and significant wave height with good accuracy. The majority of predictions lie close to the regression line, indicating a strong agreement between observed and predicted values

7 Concluding remarks

This paper presents an analysis of wave systems endemic to the Nazaré region, with particular focus on extreme wave events. The results indicate that the dominant factors contributing to wave formation vary depending on wave height, suggesting that extreme waves are governed by different mechanisms than moderate conditions. This distinction can be further exploited in machine learning applications aimed at constructing advanced wave analysis and prediction systems.

It should be noted that, although ERA5 is an advanced reanalysis product, it has limitations related to spatial resolution and its ability to accurately represent extreme events which can cause deviations.

The obtained results allow the conclusion that Nazaré represents a unique environment with respect to the occurrence of large and powerful ocean swell.

8 References

- National Geographic. *The Nazaré Canyon and Giant Waves*. Available online: <https://www.nationalgeographic.com/magazine/article/what-its-like-to-surf-the-worlds-largest-waves>
<https://www.britannica.com/place/Atlantic-Ocean/Hydrology>